

Enhanced Capture (eCAP) Module

The enhanced Capture (eCAP) module is essential in systems where accurate timing of external events is important. This microcontroller implements 6 instances of the eCAP module.

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20.1 Introduction

Uses for eCAP include:

- Speed measurements of rotating machinery (for example, toothed sprockets sensed via Hall sensors)
- Elapsed time measurements between position sensor pulses
- Period and duty cycle measurements of pulse train signals
- Decoding current or voltage amplitude derived from duty cycle encoded current/voltage sensors

The eCAP module described in this guide includes the following features:

- 4-event time-stamp registers (each 32 bits)
- Edge polarity selection for up to four sequenced time-stamp capture events
- Interrupt on either of the four events
- Single shot capture of up to four event time-stamps
- Continuous mode capture of time-stamps in a four-deep circular buffer
- Absolute time-stamp capture
- Difference (Delta) mode time-stamp capture
- All above resources dedicated to a single input pin
- When not used in capture mode, the ECAP module can be configured as a single channel PWM output

20.2 Description

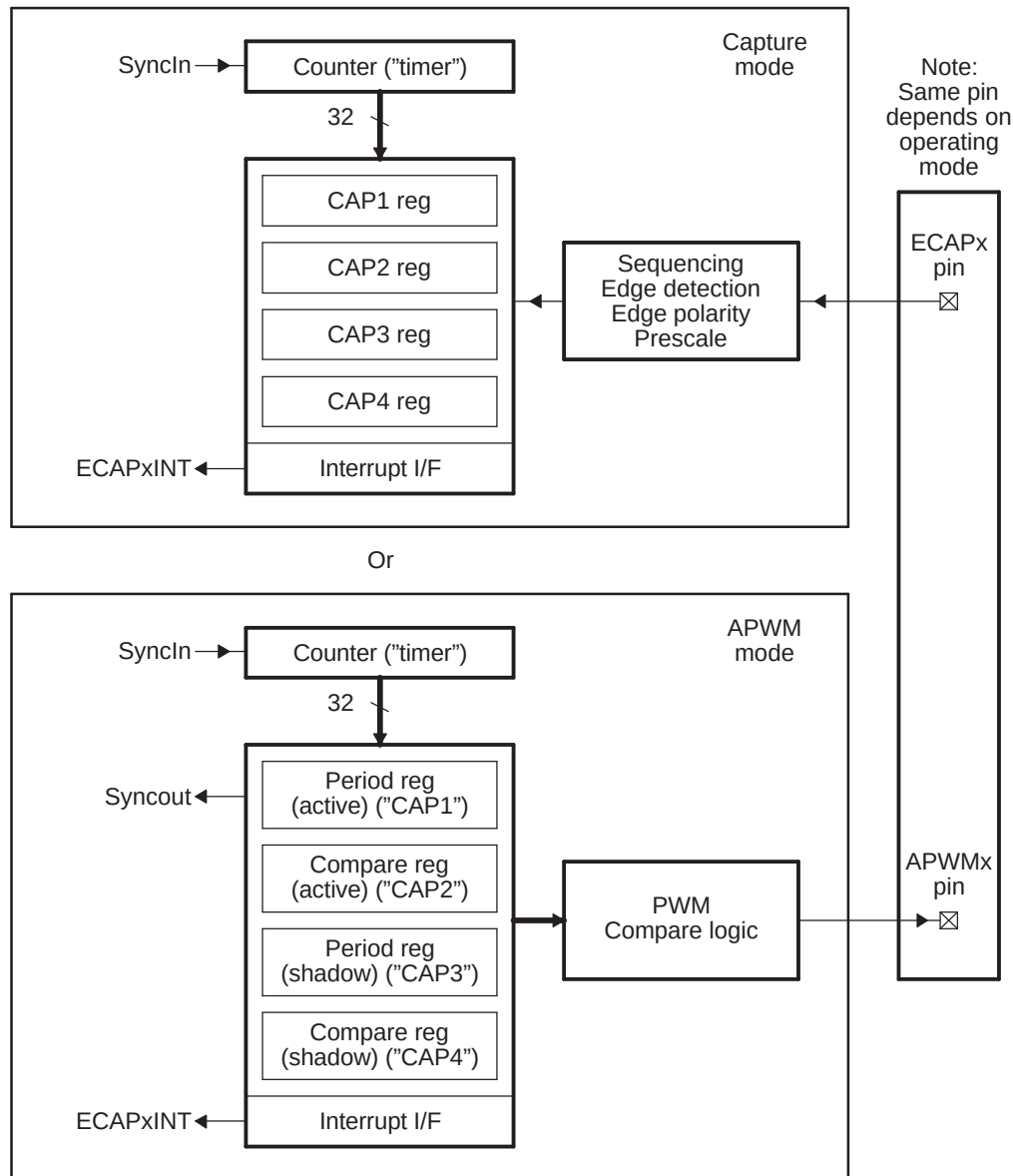
In the context of this chapter, one eCAP channel has the following independent key resources:

- Dedicated input capture pin
- 32-bit time base (counter)
- 4 × 32-bit time-stamp capture registers (CAP1-CAP4)
- 4-stage sequencer (Modulo4 counter) that is synchronized to external events, ECAP pin rising/falling edges
- Independent edge polarity (rising/falling edge) selection for all 4 events
- Input capture signal prescaling (from 2 to 62)
- One-shot compare register (2 bits) to freeze captures after 1 to 4 time-stamp events
- Control for continuous time-stamp captures using a 4-deep circular buffer (CAP1-CAP4) scheme
- Interrupt capabilities on any of the 4 capture events

20.3 Capture and APWM Operating Mode

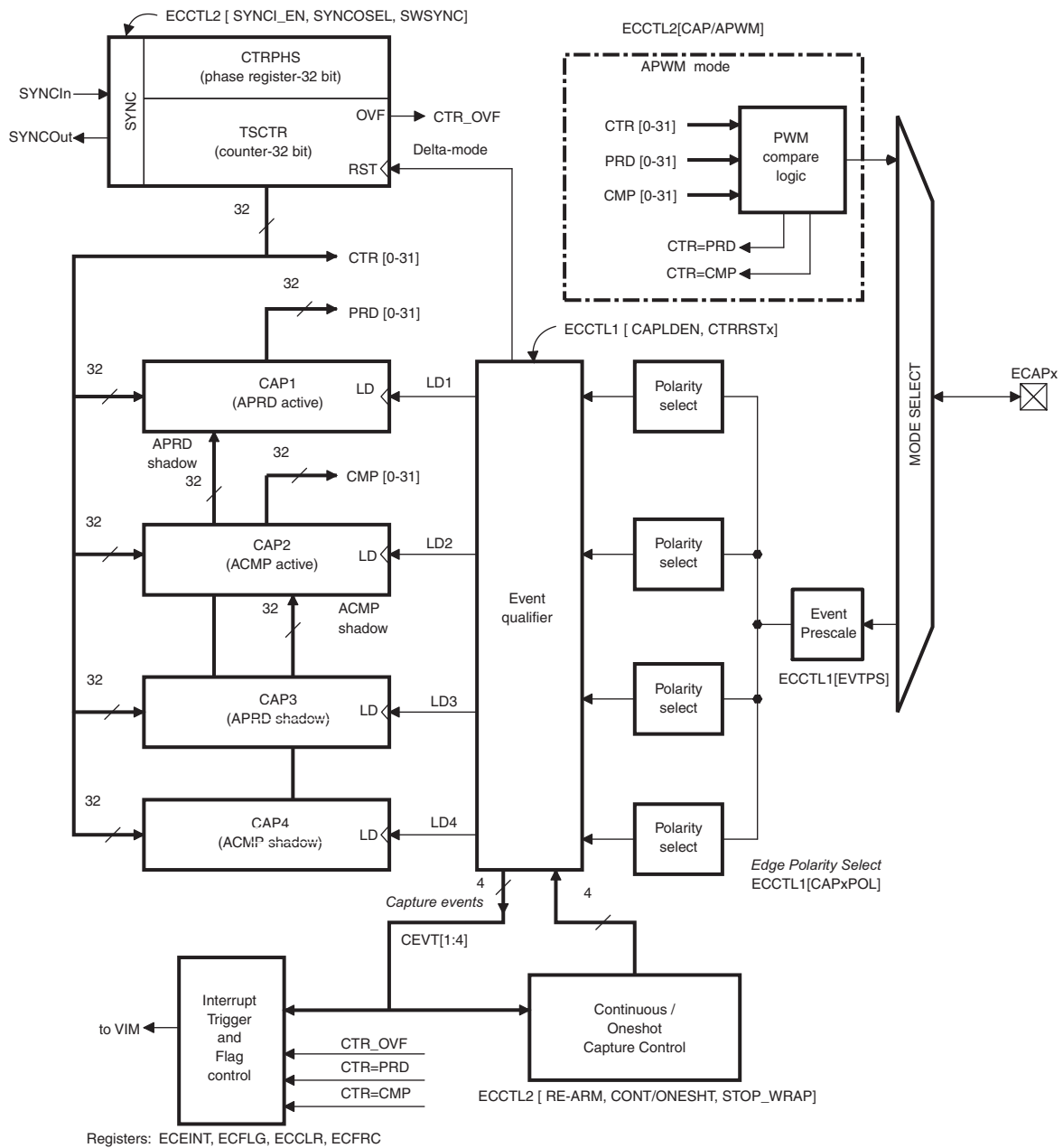
You can use the eCAP module resources to implement a single-channel PWM generator (with 32 bit capabilities) when it is not being used for input captures. The counter operates in count-up mode, providing a time-base for asymmetrical pulse width modulation (PWM) waveforms. The CAP1 and CAP2 registers become the active period and compare registers, respectively, while CAP3 and CAP4 registers become the period and capture shadow registers, respectively. Figure 20-1 is a high-level view of both the capture and auxiliary pulse-width modulator (APWM) modes of operation.

Figure 20-1. Capture and APWM Modes of Operation



- A A single pin is shared between CAP and APWM functions. In capture mode, it is an input; in APWM mode, it is an output.
- B In APWM mode, writing any value to CAP1/CAP2 active registers also writes the same value to the corresponding shadow registers CAP3/CAP4. This emulates immediate mode. Writing to the shadow registers CAP3/CAP4 invokes the shadow mode.

Figure 20-2. Capture Function Diagram

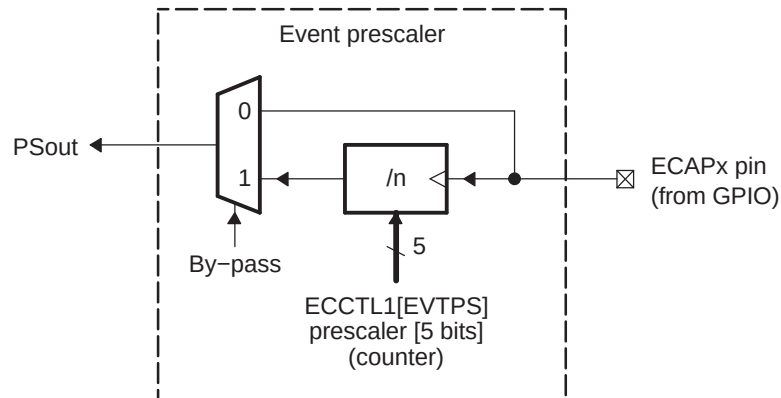


20.4.1 Event Prescaler

- An input capture signal (pulse train) can be prescaled by $N = 2-62$ (in multiples of 2) or can bypass the prescaler.

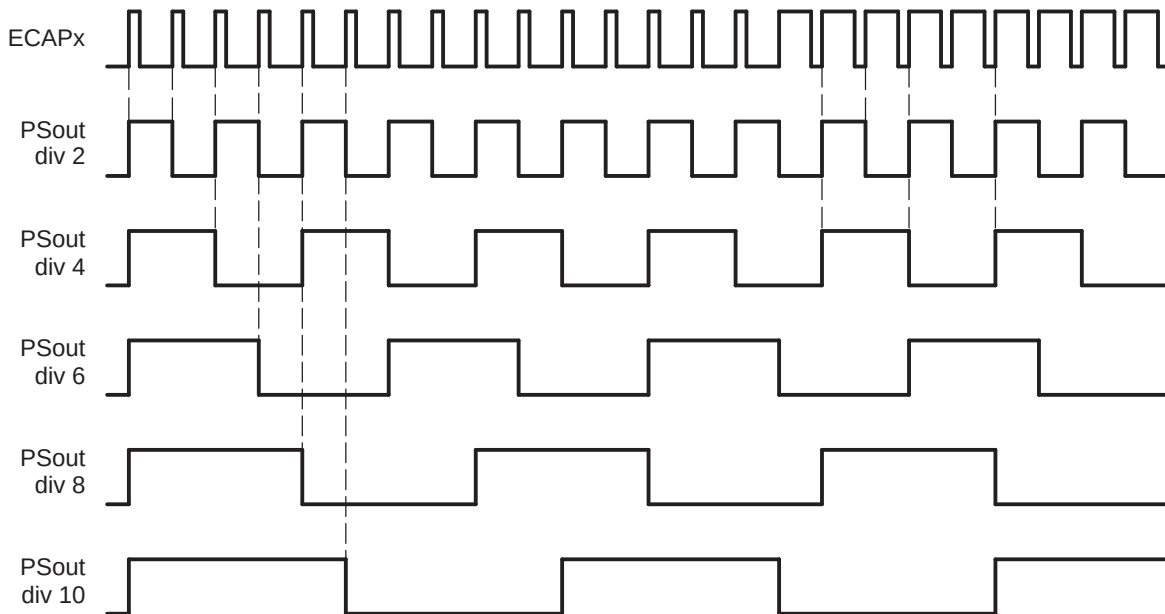
This is useful when very high frequency signals are used as inputs. [Figure 20-3](#) shows a functional diagram and [Figure 20-4](#) shows the operation of the prescale function.

Figure 20-3. Event Prescale Control



- A When a prescale value of 1 is chosen (that is, $ECCTL1[13:9] = 0,0,0,0,0$) the input capture signal by-passes the prescale logic completely.

Figure 20-4. Prescale Function Waveforms



20.4.2 Edge Polarity Select and Qualifier

- Four independent edge polarity (rising edge/falling edge) selection MUXes are used, one for each capture event.
- Each edge (up to 4) is event qualified by the Modulo4 sequencer.
- The edge event is gated to its respective CAPx register by the Mod4 counter. The CAPx register is loaded on the falling edge.

20.4.3 Continuous/One-Shot Control

- The Mod4 (2 bit) counter is incremented via edge qualified events (CEVT1-CEVT4).
- The Mod4 counter continues counting (0->1->2->3->0) and wraps around unless stopped.
- A 2-bit stop register is used to compare the Mod4 counter output, and when equal stops the Mod4 counter and inhibits further loads of the CAP1-CAP4 registers. This occurs during one-shot operation.

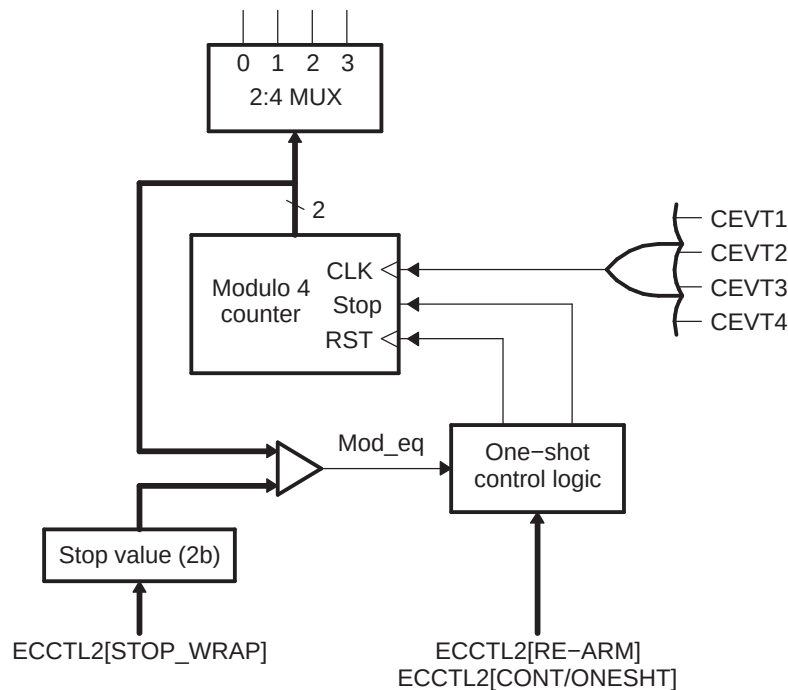
The continuous/one-shot block controls the start/stop and reset (zero) functions of the Mod4 counter via a mono-shot type of action that can be triggered by the stop-value comparator and re-armed via software control.

Once armed, the eCAP module waits for 1-4 (defined by stop-value) capture events before freezing both the Mod4 counter and contents of CAP1-4 registers (that is, time-stamps).

Re-arming prepares the eCAP module for another capture sequence. Also re-arming clears (to zero) the Mod4 counter and permits loading of CAP1-4 registers again, providing the CAPLDEN bit is set.

In continuous mode, the Mod4 counter continues to run (0->1->2->3->0, the one-shot action is ignored, and capture values continue to be written to CAP1-4 in a circular buffer sequence.

Figure 20-5. Details of the Continuous/One-shot Block



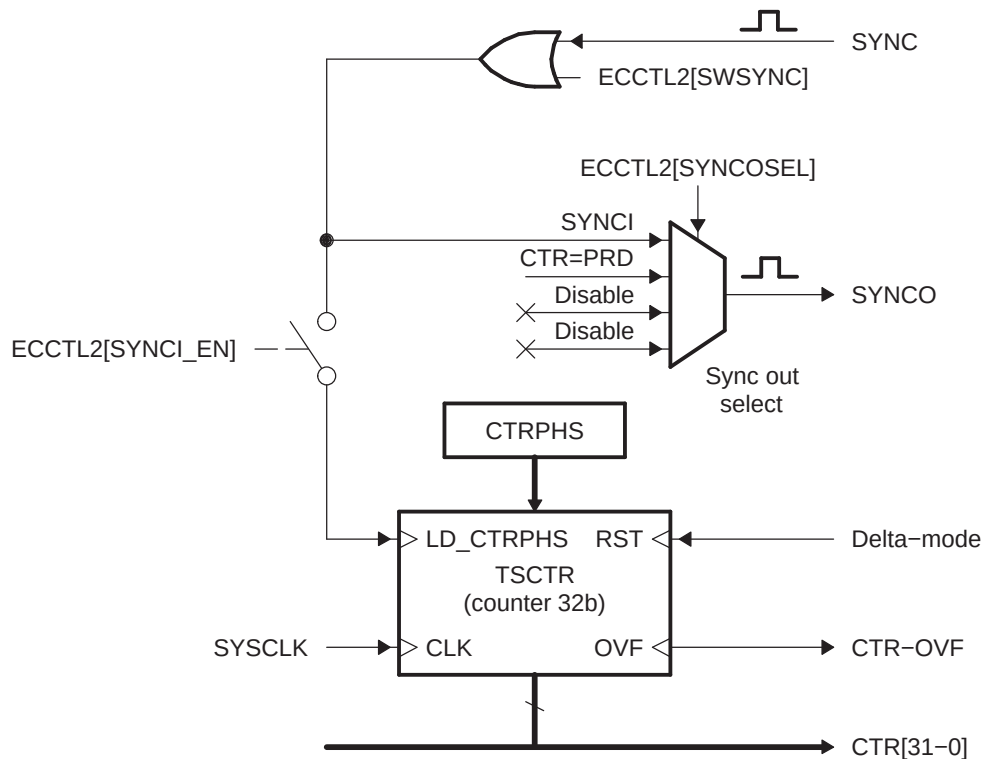
20.4.4 32-Bit Counter and Phase Control

This counter provides the time-base for event captures, and is clocked via the system clock.

A phase register is provided to achieve synchronization with other counters, via a hardware and software forced sync. This is useful in APWM mode when a phase offset between modules is needed.

On any of the four event loads, an option to reset the 32-bit counter is given. This is useful for time difference capture. The 32-bit counter value is captured first, then it is reset to 0 by any of the LD1-LD4 signals.

Figure 20-6. Details of the Counter and Synchronization Block



20.4.5 CAP1-CAP4 Registers

These 32-bit registers are fed by the 32-bit counter timer bus, CTR[0-31] and are loaded (that is, capture a time-stamp) when their respective LD inputs are strobed.

Loading of the capture registers can be inhibited via control bit CAPLDEN. During one-shot operation, this bit is cleared (loading is inhibited) automatically when a stop condition occurs, that is, StopValue = Mod4.

CAP1 and CAP2 registers become the active period and compare registers, respectively, in APWM mode.

CAP3 and CAP4 registers become the respective shadow registers (APRD and ACMP) for CAP1 and CAP2 during APWM operation.

20.4.6 Interrupt Control

An interrupt can be generated on capture events (CEVT1-CEVT4, CTROVF) or APWM events (CTR = PRD, CTR = CMP).

A counter overflow event (FFFFFFFF->00000000) is also provided as an interrupt source (CTROVF).

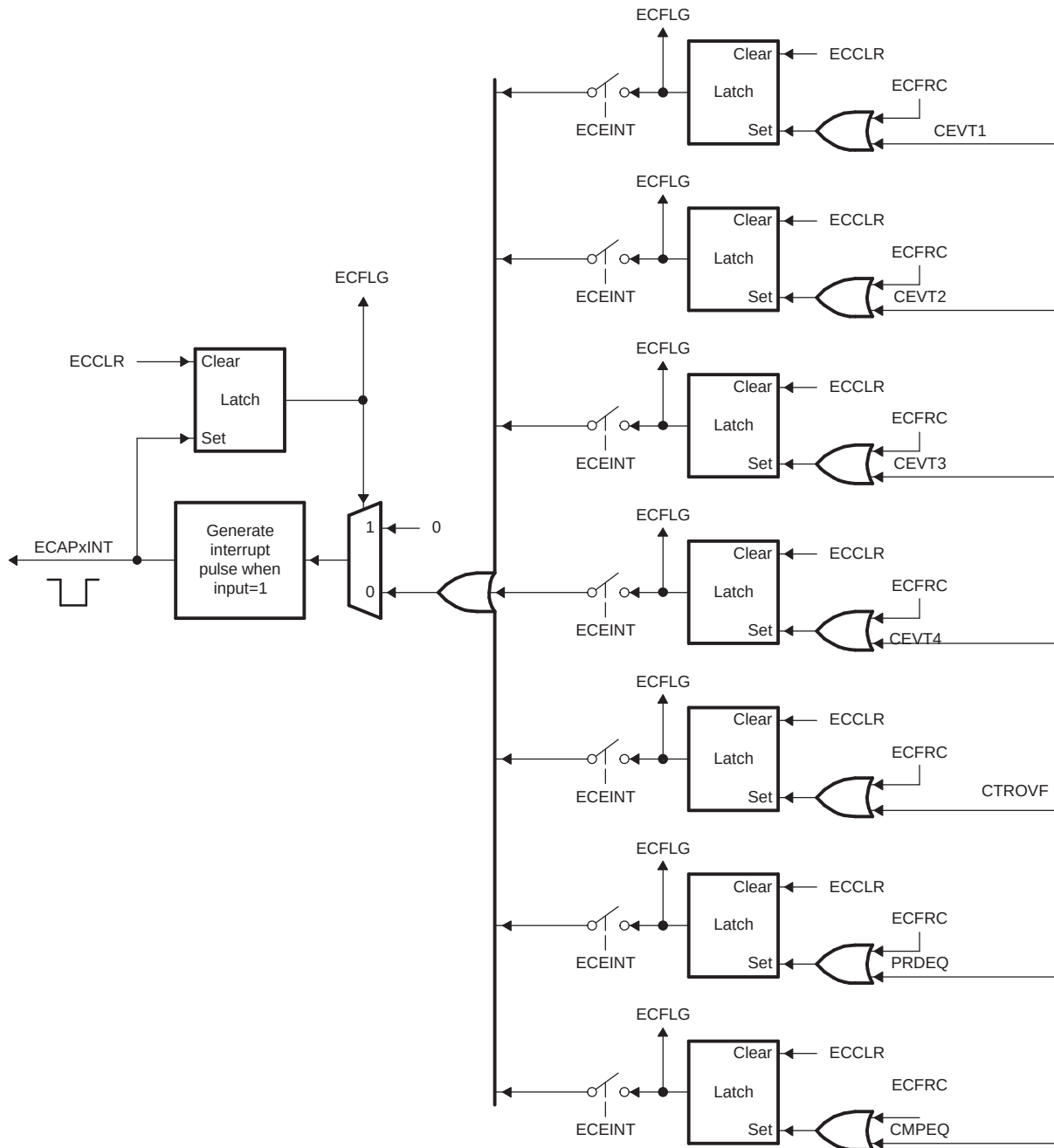
The capture events are edge and sequencer qualified (that is, ordered in time) by the polarity select and Mod4 gating, respectively.

One of these events can be selected as the interrupt source (from the eCAPx module) going to the PIE.

Seven interrupt events (CEVT1, CEVT2, CEVT3, CEVT4, CNTOVF, CTR=PRD, CTR=CMP) can be generated. The interrupt enable register (ECEINT) is used to enable/disable individual interrupt event sources. The interrupt flag register (ECFLG) indicates if any interrupt event has been latched and contains the global interrupt flag bit (INT). An interrupt pulse is generated to the PIE only if any of the interrupt events are enabled, the flag bit is 1, and the INT flag bit is 0. The interrupt service routine must clear the global interrupt flag bit and the serviced event via the interrupt clear register (ECCLR) before any other interrupt pulses are generated. You can force an interrupt event via the interrupt force register (ECFRC). This is useful for test purposes.

Note: The CEVT1, CEVT2, CEVT3, CEVT4 flags are only active in capture mode (ECCTL2[CAP_APWM == 0]). The CTR_PRD, CTR_CMP flags are only valid in APWM mode (ECCTL2[CAP_APWM == 1]). CNTOVF flag is valid in both modes.

Figure 20-7. Interrupts in eCAP Module



20.4.7 Shadow Load and Lockout Control

In capture mode, this logic inhibits (locks out) any shadow loading of CAP1 or CAP2 from APRD and ACMP registers, respectively.

In APWM mode, shadow loading is active and two choices are permitted:

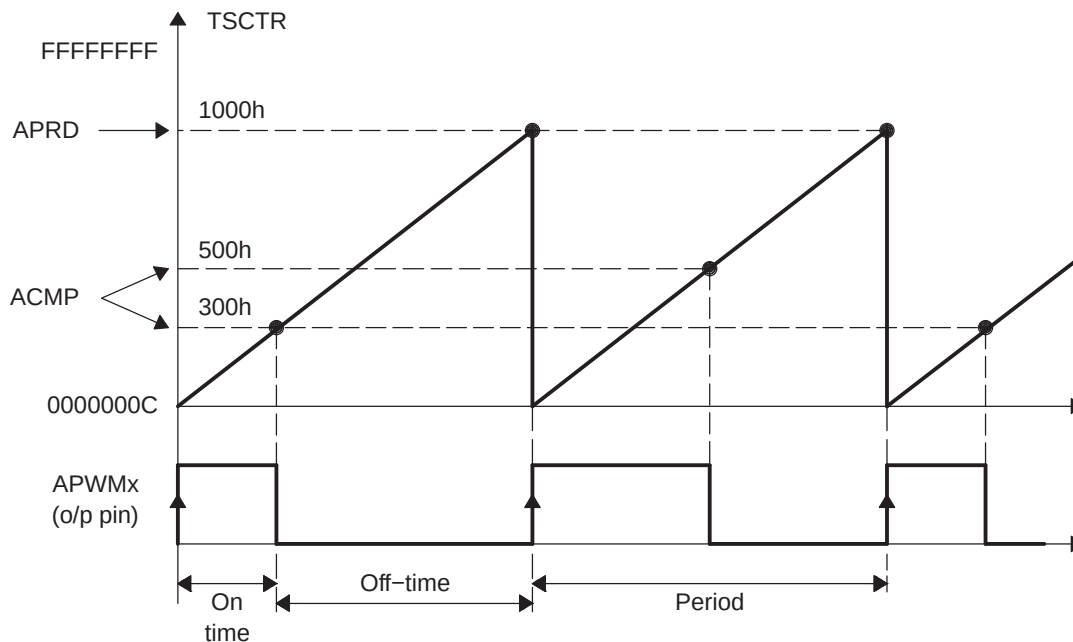
- Immediate - APRD or ACMP are transferred to CAP1 or CAP2 immediately upon writing a new value.
- On period equal, that is, CTR[31:0] = PRD[31:0]

20.4.8 APWM Mode Operation

Main operating highlights of the APWM section:

- The time-stamp counter bus is made available for comparison via 2 digital (32-bit) comparators.
- When CAP1/2 registers are not used in capture mode, their contents can be used as Period and Compare values in APWM mode.
- Double buffering is achieved via shadow registers APRD and ACMP (CAP3/4). The shadow register contents are transferred over to CAP1/2 registers either immediately upon a write, or on a CTR = PRD trigger.
- In APWM mode, writing to CAP1/CAP2 active registers will also write the same value to the corresponding shadow registers CAP3/CAP4. This emulates immediate mode. Writing to the shadow registers CAP3/CAP4 will invoke the shadow mode.
- During initialization, you must write to the active registers for both period and compare. This automatically copies the initial values into the shadow values. For subsequent compare updates, that is, during run-time, you only need to use the shadow registers.

Figure 20-8. PWM Waveform Details Of APWM Mode Operation



The behavior of APWM active high mode (APWMPOL == 0) is as follows:

CMP = 0x00000000, output low for duration of period (0% duty)

CMP = 0x00000001, output high 1 cycle

CMP = 0x00000002, output high 2 cycles

CMP = PERIOD, output high except for 1 cycle (<100% duty)

CMP = PERIOD+1, output high for complete period (100% duty)

CMP > PERIOD+1, output high for complete period

The behavior of APWM active low mode (APWMPOL == 1) is as follows:

CMP = 0x00000000, output high for duration of period (0% duty)

CMP = 0x00000001, output low 1 cycle

CMP = 0x00000002, output low 2 cycles

CMP = PERIOD, output low except for 1 cycle (<100% duty)

CMP = PERIOD+1, output low for complete period (100% duty)

CMP > PERIOD+1, output low for complete period

20.5 ECAP Registers

[Table 20-1](#) lists the memory-mapped registers for the ECAP. All register offset addresses not listed in [Table 20-1](#) should be considered as reserved locations and the register contents should not be modified.

The base address for the control registers is FCF7 9300h for eCAP1, FCF7 9400h for eCAP2, FCF7 9500h for eCAP3, FCF7 9600h for eCAP4, FCF7 9700h for eCAP5, and FCF7 9800h for eCAP6.

Table 20-1. ECAP Registers

Offset	Acronym	Register Name	Section
0h	TSCTR	Time-Stamp Counter	Section 20.5.1
4h	CTRPHS	Counter Phase Offset Value Register	Section 20.5.2
8h	CAP1	Capture 1 Register	Section 20.5.3
Ch	CAP2	Capture 2 Register	Section 20.5.4
10h	CAP3	Capture 3 Register	Section 20.5.5
14h	CAP4	Capture 4 Register	Section 20.5.6
28h	ECCTL1	Capture Control Register 1	Section 20.5.7
2Ah	ECCTL2	Capture Control Register 2	Section 20.5.8
2Ch	ECEINT	Capture Interrupt Enable Register	Section 20.5.9
2Eh	ECFLG	Capture Interrupt Flag Register	Section 20.5.10
30h	ECCLR	Capture Interrupt Clear Register	Section 20.5.11
32h	ECFRC	Capture Interrupt Force Register	Section 20.5.12

Complex bit access types are encoded to fit into small table cells. [Table 20-2](#) shows the codes that are used for access types in this section.

Table 20-2. ECAP Access Type Codes

Access Type	Code	Description
Read Type		
R	R	Read
Write Type		
W	W	Write
Reset or Default Value		
-n		Value after reset or the default value

20.5.1 TSCTR Register (Offset = 0h) [reset = 0h]

TSCTR is shown in [Figure 20-9](#) and described in [Table 20-3](#).

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Figure 20-9. TSCTR Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TSCTR																															
R/W-0h																															

Table 20-3. TSCTR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	TSCTR	R/W	0h	Active 32-bit counter register that is used as the capture time-base.

20.5.2 CTRPHS Register (Offset = 4h) [reset = 0h]

CTRPHS is shown in [Figure 20-10](#) and described in [Table 20-4](#).

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Figure 20-10. CTRPHS Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CTRPHS																															
R/W-0h																															

Table 20-4. CTRPHS Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	CTRPHS	R/W	0h	Counter phase value register that can be programmed for phase lag/lead. This register shadows TSCTR and is loaded into TSCTR upon either a SYNCI event or software force via a control bit. Used to achieve phase control synchronization with respect to other eCAP and ePWM time-bases.

20.5.3 CAP1 Register (Offset = 8h) [reset = 0h]

CAP1 is shown in [Figure 20-11](#) and described in [Table 20-5](#).

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In APWM mode, writing to CAP1/CAP2 active registers also writes the same value to the corresponding shadow registers CAP3/CAP4. This emulates immediate mode. Writing to the shadow registers CAP3/CAP4 invokes the shadow mode.

Figure 20-11. CAP1 Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CAP1																															
R/W-0h																															

Table 20-5. CAP1 Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	CAP1	R/W	0h	This register can be loaded (written) by: Time-Stamp (that is, counter value) during a capture event. Software - may be useful for test purposes. APRD shadow register (that is, CAP4) when used in APWM mode.

20.5.4 CAP2 Register (Offset = Ch) [reset = 0h]

CAP2 is shown in [Figure 20-12](#) and described in [Table 20-6](#).

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In APWM mode, writing to CAP1/CAP2 active registers also writes the same value to the corresponding shadow registers CAP3/CAP4. This emulates immediate mode. Writing to the shadow registers CAP3/CAP4 invokes the shadow mode.

Figure 20-12. CAP2 Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CAP2																															
R/W-0h																															

Table 20-6. CAP2 Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	CAP2	R/W	0h	This register can be loaded (written) by: Time-Stamp (that is, counter value) during a capture event. Software - may be useful for test purposes. APRD shadow register (that is, CAP4) when used in APWM mode.

20.5.5 CAP3 Register (Offset = 10h) [reset = 0h]

CAP3 is shown in [Figure 20-13](#) and described in [Table 20-7](#).

Return to [Summary Table](#).

Figure 20-13. CAP3 Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CAP3																															
R/W-0h																															

Table 20-7. CAP3 Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	CAP3	R/W	0h	In CMP mode, this is a time-stamp capture register. In APWM mode, this is the period shadow (APRD) register. Update the PWM period value through this register. In this mode, CAP3 (APRD) shadows CAP1.

20.5.6 CAP4 Register (Offset = 14h) [reset = 0h]

CAP4 is shown in [Figure 20-14](#) and described in [Table 20-8](#).

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Figure 20-14. CAP4 Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CAP4																															
R/W-0h																															

Table 20-8. CAP4 Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	CAP4	R/W	0h	In CMP mode, this is a time-stamp capture register. In APWM mode, this is the compare shadow (ACMP) register. Update the PWM compare value via this register. In this mode, CAP4 (ACMP) shadows CAP2.

20.5.7 ECCTL1 Register (Offset = 28h) [reset = 0h]

ECCTL1 is shown in [Figure 20-15](#) and described in [Table 20-9](#).

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Figure 20-15. ECCTL1 Register

15	14	13	12	11	10	9	8
FREE_SOFT		PRESCALE					CAPLDEN
R/W-0h		R/W-0h					R/W-0h
7	6	5	4	3	2	1	0
CTRRST4	CAP4POL	CTRRST3	CAP3POL	CTRRST2	CAP2POL	CTRRST1	CAP1POL
R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h

Table 20-9. ECCTL1 Register Field Descriptions

Bit	Field	Type	Reset	Description
15-14	FREE_SOFT	R/W	0h	Emulation Control. 0h = TSCTR counter stops immediately on emulation suspend. 1h = TSCTR counter runs until value equals 0. 2h = TSCTR counter is unaffected by emulation suspend (Run Free). 3h = TSCTR counter is unaffected by emulation suspend (Run Free).
13-9	PRESCALE	R/W	0h	Event Filter prescale select. Valid values: 0 to 1Fh. 0h = Divide by 1 (that is, no prescale, by-pass the prescaler) 1h = Divide by 2 2h = Divide by 4 3h = Divide by 6 4h = Divide by 8 5h = Divide by 10 1Eh = Divide by 60 1Fh = Divide by 62
8	CAPLDEN	R/W	0h	Enable Loading of CAP1-4 registers on a capture event. 0h = Disable CAP1-4 register loads at capture event time. 1h = Enable CAP1-4 register loads at capture event time.
7	CTRRST4	R/W	0h	Counter Reset on Capture Event 4. 0h = Do not reset counter on Capture Event 4 (absolute time stamp operation). 1h = Reset counter after Capture Event 4 time-stamp has been captured (used in difference mode operation).
6	CAP4POL	R/W	0h	Capture Event 4 Polarity select. 0h = Capture Event 4 triggered on a rising edge (RE). 1h = Capture Event 4 triggered on a falling edge (FE).
5	CTRRST3	R/W	0h	Counter Reset on Capture Event 3. 0h = Do not reset counter on Capture Event 3 (absolute time stamp). 1h = Reset counter after Event 3 time-stamp has been captured (used in difference mode operation).
4	CAP3POL	R/W	0h	Capture Event 3 Polarity select. 0h = Capture Event 3 triggered on a rising edge (RE). 1h = Capture Event 3 triggered on a falling edge (FE).
3	CTRRST2	R/W	0h	Counter Reset on Capture Event 2. 0h = Do not reset counter on Capture Event 2 (absolute time stamp). 1h = Reset counter after Event 2 time-stamp has been captured (used in difference mode operation).

Table 20-9. ECCTL1 Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
2	CAP2POL	R/W	0h	Capture Event 2 Polarity select. 0h = Capture Event 2 triggered on a rising edge (RE). 1h = Capture Event 2 triggered on a falling edge (FE).
1	CTRRST1	R/W	0h	Counter Reset on Capture Event 1. 0h = Do not reset counter on Capture Event 1 (absolute time stamp). 1h = Reset counter after Event 1 time-stamp has been captured (used in difference mode operation).
0	CAP1POL	R/W	0h	Capture Event 1 Polarity select. 0h = Capture Event 1 triggered on a rising edge (RE). 1h = Capture Event 1 triggered on a falling edge (FE).

20.5.8 ECCTL2 Register (Offset = 2Ah) [reset = 2h]

ECCTL2 is shown in [Figure 20-16](#) and described in [Table 20-10](#).

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Figure 20-16. ECCTL2 Register

15	14	13	12	11	10	9	8
RESERVED					APWMPOL	CAP_APWM	SWSYNC
R-0h					R/W-0h	R/W-0h	R/W-0h
7	6	5	4	3	2	1	0
SYNCO_SEL	SYNCl_EN		TSCTRSTOP	REARM	STOP_WRAP		CONT_ONESHOT
R/W-0h	R/W-0h		R/W-0h	R/W-0h	R/W-1h		R/W-0h

Table 20-10. ECCTL2 Register Field Descriptions

Bit	Field	Type	Reset	Description
15-11	RESERVED	R	0h	Reserved
10	APWMPOL	R/W	0h	APWM output polarity select. This is applicable only in APWM operating mode. 0h = Output is active high (that is, Compare value defines high time). 1h = Output is active low (that is, Compare value defines low time).
9	CAP_APWM	R/W	0h	CAP/APWM operating mode select. 0h = ECAP module operates in capture mode. This mode forces the following configuration: inhibits TSCTR resets via CTR = PRD event, inhibits shadow loads on CAP1 and CAP2 registers, permits user to enable CAP1 to CAP4 register load, CAPx and APWMx pin operates as a capture input. 1h = ECAP module operates in APWM mode. This mode forces the following configuration: resets TSCTR on CTR = PRD event (period boundary, permits shadow loading on CAP1 and CAP2 registers, disables loading of time-stamps into CAP1 to CAP4 registers, CAPx and APWMx pin operates as a APWM output.
8	SWSYNC	R/W	0h	Software-forced Counter (TSCTR) Synchronizing. This provides a convenient software method to synchronize some or all ECAP time bases. In APWM mode, the synchronizing can also be done via the CTR = PRD event. Note: Selection CTR = PRD is meaningful only in APWM mode, however, you can choose it in CAP mode if you find doing so is useful. 0h = Writing a 0 has no effect. Reading always returns 0. 1h = Writing a 1 forces a TSCTR shadow load of current ECAP module and any ECAP modules down-stream providing the SYNCO_SEL bits are 0,0. After writing a 1, this bit returns to 0.
7-6	SYNCO_SEL	R/W	0h	Sync-Out Select. 0h = Select sync-in event to be the sync-out signal (pass through). 1h = Select CTR = PRD event to be the sync-out signal. 2h = Disable sync out signal. 3h = Disable sync out signal.
5	SYNCl_EN	R/W	0h	Counter (TSCTR) Sync-In select mode. 0h = Disable sync-in option. 1h = Enable counter (TSCTR) to be loaded from CTRPHS register upon either a SYNCl signal or a software force event.
4	TSCTRSTOP	R/W	0h	Time Stamp (TSCTR) Counter Stop (freeze) Control. 0h = TSCTR is stopped. 1h = TSCTR is free-running.

Table 20-10. ECCTL2 Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
3	REARM	R/W	0h	One-Shot Re-Arming Control, that is, wait for stop trigger. Note: The re-arm function is valid in one shot or continuous mode. 0h = Writing a 0 has no effect. Reading always returns 0. 1h = Arms the one-shot sequence as follows: resets the Mod4 counter to 0, unfreezes the Mod4 counter, enables capture register loads.
2-1	STOP_WRAP	R/W	1h	Stop value for one-shot mode. This is the number (between 1-4) of captures allowed to occur before the CAP(1-4) registers are frozen, that is, capture sequence is stopped. Notes: STOP_WRAP is compared to Mod4 counter and, when equal, two actions occur: 1) Mod4 counter is stopped (frozen). 2) Capture register loads are inhibited. In one-shot mode, further interrupt events are blocked until re-armed. Wrap value for continuous mode. This is the number (between 1-4) of the capture register in which the circular buffer wraps around and starts again. 0h = Stop after Capture Event 1 in one-shot mode. Wrap after Capture Event 1 in continuous mode. 1h = Stop after Capture Event 2 in one-shot mode. Wrap after Capture Event 2 in continuous mode. 2h = Stop after Capture Event 3 in one-shot mode. Wrap after Capture Event 3 in continuous mode. 3h = Stop after Capture Event 4 in one-shot mode. Wrap after Capture Event 4 in continuous mode.
0	CONT_ONESHT	R/W	0h	Continuous or one-shot mode control. This is applicable only in capture mode. 0h = Operate in continuous mode. 1h = Operate in one-shot mode.

20.5.9 ECEINT Register (Offset = 2Ch) [reset = 0h]

ECEINT is shown in [Figure 20-17](#) and described in [Table 20-11](#).

Return to [Summary Table](#).

The interrupt enable bits (CEVT1, ...) block any of the selected events from generating an interrupt. Events are still latched into the flag bit (ECFLG register) and can be forced/cleared via the ECFRC/ECCLR registers.

The proper procedure for configuring peripheral modes and interrupts is as follows:

- 1) Disable global interrupts.
- 2) Stop eCAP counter.
- 3) Disable eCAP interrupts.
- 4) Configure peripheral registers.
- 5) Clear spurious eCAP interrupt flags.
- 6) Enable eCAP interrupts.
- 7) Start eCAP counter.
- 8) Enable global interrupts.

Figure 20-17. ECEINT Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
CTR_CMP	CTR_PRD	CTROVF	CEVT4	CEVT3	CEVT2	CEVT1	RESERVED
R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R-0h

Table 20-11. ECEINT Register Field Descriptions

Bit	Field	Type	Reset	Description
15-8	RESERVED	R	0h	Reserved
7	CTR_CMP	R/W	0h	Counter Equal Compare Interrupt Enable. 0h = Disable Compare Equal as an Interrupt source. 1h = Enable Compare Equal as an Interrupt source.
6	CTR_PRD	R/W	0h	Counter Equal Period Interrupt Enable. 0h = Disable Period Equal as an Interrupt source. 1h = Enable Period Equal as an Interrupt source.
5	CTROVF	R/W	0h	Counter Overflow Interrupt Enable. 0h = Disabled Counter Overflow as an Interrupt source. 1h = Enable Counter Overflow as an Interrupt source.
4	CEVT4	R/W	0h	Capture Event 4 Interrupt Enable. 0h = Disable Capture Event 4 as an Interrupt source. 1h = Enable Capture Event 4 as an Interrupt source.
3	CEVT3	R/W	0h	Capture Event 3 Interrupt Enable. 0h = Disable Capture Event 3 as an Interrupt source. 1h = Enable Capture Event 3 as an Interrupt source.
2	CEVT2	R/W	0h	Capture Event 2 Interrupt Enable. 0h = Disable Capture Event 2 as an Interrupt source. 1h = Enable Capture Event 2 as an Interrupt source.
1	CEVT1	R/W	0h	Capture Event 1 Interrupt Enable. 0h = Disable Capture Event 1 as an Interrupt source. 1h = Enable Capture Event 1 as an Interrupt source.
0	RESERVED	R	0h	Reserved

20.5.10 ECFLG Register (Offset = 2Eh) [reset = 0h]

ECFLG is shown in [Figure 20-18](#) and described in [Table 20-12](#).

Return to [Summary Table](#).

Figure 20-18. ECFLG Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
CTR_CMP	CTR_PRD	CTROVF	CEVT4	CEVT3	CEVT2	CEVT1	INT
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h

Table 20-12. ECFLG Register Field Descriptions

Bit	Field	Type	Reset	Description
15-8	RESERVED	R	0h	Reserved
7	CTR_CMP	R	0h	Compare Equal Compare Status Flag. This flag is active only in APWM mode. 0h = No event occurred. 1h = The counter (TSCTR) reached the compare register value (ACMP).
6	CTR_PRD	R	0h	Counter Equal Period Status Flag. This flag is only active in APWM mode. 0h = No event occurred. 1h = The counter (TSCTR) reached the period register value (APRD) and was reset.
5	CTROVF	R	0h	Counter Overflow Status Flag. This flag is active in CAP and APWM mode. 0h = No event occurred. 1h = The counter (TSCTR) has made the transition from FFFFFFFF to 00000000.
4	CEVT4	R	0h	Capture Event 4 Status Flag This flag is only active in CAP mode. 0h = No event occurred. 1h = The fourth event occurred at ECAPx pin.
3	CEVT3	R	0h	Capture Event 3 Status Flag. This flag is active only in CAP mode. 0h = No event occurred. 1h = The third event occurred at ECAPx pin.
2	CEVT2	R	0h	Capture Event 2 Status Flag. This flag is only active in CAP mode. 0h = No event occurred. 1h = The second event occurred at ECAPx pin.
1	CEVT1	R	0h	Capture Event 1 Status Flag. This flag is only active in CAP mode. 0h = No event occurred. 1h = The first event occurred at ECAPx pin.
0	INT	R	0h	Global Interrupt Status Flag. 0h = No interrupt is generated. 1h = An interrupt is generated.

20.5.11 ECCLR Register (Offset = 30h) [reset = 0h]

ECCLR is shown in [Figure 20-19](#) and described in [Table 20-13](#).

[Return to Summary Table.](#)

Figure 20-19. ECCLR Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
CTR_CMP	CTR_PRD	CTROVF	CEVT4	CEVT3	CEVT2	CEVT1	INT
R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h

Table 20-13. ECCLR Register Field Descriptions

Bit	Field	Type	Reset	Description
15-8	RESERVED	R	0h	Any writes to these bits must always have a value of 0.
7	CTR_CMP	R/W	0h	Counter Equal Compare Status Flag. 0h = Writing a 0 has no effect. Reading always returns 0. 1h = Writing a 1 clears the CTR_CMP flag condition.
6	CTR_PRD	R/W	0h	Counter Equal Period Status Flag. 0h = Writing a 0 has no effect. Reading always returns 0. 1h = Writing a 1 clears the CTR_PRD flag condition.
5	CTROVF	R/W	0h	Counter Overflow Status Flag. 0h = Writing a 0 has no effect. Reading always returns 0. 1h = Writing a 1 clears the CTROVF flag condition.
4	CEVT4	R/W	0h	Capture Event 4 Status Flag. 0h = Writing a 0 has no effect. Reading always returns 0. 1h = Writing a 1 clears the CEVT4 flag condition.
3	CEVT3	R/W	0h	Capture Event 3 Status Flag. 0h = Writing a 0 has no effect. Reading always returns 0. 1h = Writing a 1 clears the CEVT3 flag condition.
2	CEVT2	R/W	0h	Capture Event 2 Status Flag. 0h = Writing a 0 has no effect. Reading always returns 0. 1h = Writing a 1 clears the CEVT2 flag condition.
1	CEVT1	R/W	0h	Capture Event 1 Status Flag. 0h = Writing a 0 has no effect. Reading always returns 0. 1h = Writing a 1 clears the CEVT1 flag condition.
0	INT	R/W	0h	Global Interrupt Clear Flag. 0h = Writing a 0 has no effect. Reading always returns 0. 1h = Writing a 1 clears the INT flag and enable further interrupts to be generated if any of the event flags are set to 1.

20.5.12 ECFRC Register (Offset = 32h) [reset = 0h]

ECFRC is shown in [Figure 20-20](#) and described in [Table 20-14](#).

Return to [Summary Table](#).

Figure 20-20. ECFRC Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
CTR_CMP	CTR_PRD	CTROVF	CEVT4	CEVT3	CEVT2	CEVT1	RESERVED
R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R-0h

Table 20-14. ECFRC Register Field Descriptions

Bit	Field	Type	Reset	Description
15-8	RESERVED	R	0h	Any writes to these bits must always have a value of 0.
7	CTR_CMP	R/W	0h	Force Counter Equal Compare Interrupt. 0h = Writing a 0 has no effect. Reading always returns 0. 1h = Writing a 1 sets the CTR_CMP flag bit.
6	CTR_PRD	R/W	0h	Force Counter Equal Period Interrupt. 0h = Writing a 0 has no effect. Reading always returns 0. 1h = Writing a 1 sets the CTR_PRD flag bit.
5	CTROVF	R/W	0h	Force Counter Overflow. 0h = Writing a 0 has no effect. Reading always returns 0. 1h = Writing a 1 to this bit sets the CTROVF flag bit.
4	CEVT4	R/W	0h	Force Capture Event 4. 0h = Writing a 0 has no effect. Reading always returns 0. 1h = Writing a 1 sets the CEVT4 flag bit
3	CEVT3	R/W	0h	Force Capture Event 3. 0h = Writing a 0 has no effect. Reading always returns 0. 1h = Writing a 1 sets the CEVT3 flag bit
2	CEVT2	R/W	0h	Force Capture Event 2. 0h = Writing a 0 has no effect. Reading always returns 0. 1h = Writing a 1 sets the CEVT2 flag bit.
1	CEVT1	R/W	0h	Force Capture Event 1. 0h = Writing a 0 has no effect. Reading always returns 0. 1h = Sets the CEVT1 flag bit.
0	RESERVED	R	0h	Any writes to these bits must always have a value of 0.

20.6 Application of the ECAP Module

The following sections will provide applications examples and code snippets to show how to configure and operate the eCAP module. For clarity and ease of use, the examples use the eCAP “C” header files. Below are useful #defines that will help in the understanding of the examples.

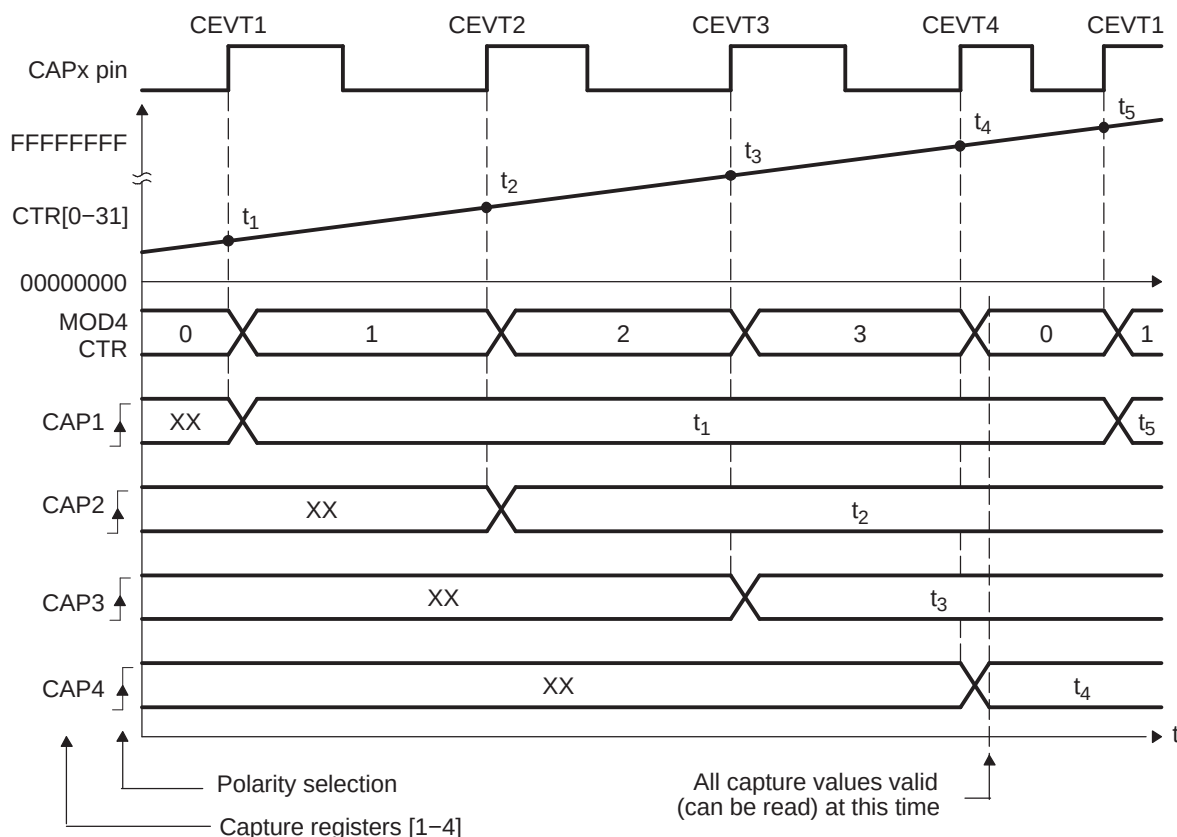
```
// ECCTL1 ( ECAP Control Reg 1)
//=====
// CAPxPOL bits
#define EC_RISING 0x0
#define EC_FALLING 0x1
// CTRRSTx bits
#define EC_ABS_MODE 0x0
#define EC_DELTA_MODE 0x1
// PRESCALE bits
#define EC_BYPASS 0x0
#define EC_DIV1 0x0
#define EC_DIV2 0x1
#define EC_DIV4 0x2
#define EC_DIV6 0x3
#define EC_DIV8 0x4
#define EC_DIV10 0x5
// ECCTL2 ( ECAP Control Reg 2)
//=====
// CONT/ONESHOT bit
#define EC_CONTINUOUS 0x0
#define EC_ONESHOT 0x1
// STOPVALUE bit
#define EC_EVENT1 0x0
#define EC_EVENT2 0x1
#define EC_EVENT3 0x2
#define EC_EVENT4 0x3
// RE-ARM bit
#define EC_ARM 0x1
// TSCTRSTOP bit
#define EC_FREEZE 0x0
#define EC_RUN 0x1
// SYNC0_SEL bit
#define EC_SYNCIN 0x0
#define EC_CTR_PRD 0x1
#define EC_SYNC0_DIS 0x2
// CAP/APWM mode bit
#define EC_CAP_MODE 0x0
#define EC_APWM_MODE 0x1
// APWMPOL bit
#define EC_ACTV_HI 0x0
#define EC_ACTV_LO 0x1
// Generic
#define EC_DISABLE 0x0
#define EC_ENABLE 0x1
#define EC_FORCE 0x1
```

20.6.1 Example 1 - Absolute Time-Stamp Operation Rising Edge Trigger

Figure 20-21 shows an example of continuous capture operation (Mod4 counter wraps around). In this figure, TSCTR counts-up without resetting and capture events are qualified on the rising edge only, this gives period (and frequency) information.

On an event, the TSCTR contents (that is, time-stamp) is first captured, then Mod4 counter is incremented to the next state. When the TSCTR reaches FFFF FFFFh (that is, maximum value), it wraps around to 0000 0000h (not shown in Figure 20-21), if this occurs, the CTROVF (counter overflow) flag is set, and an interrupt (if enabled) occurs, CTROVF (counter overflow) Flag is set, and an Interrupt (if enabled) occurs. Captured Time-stamps are valid at the point indicated by the diagram, that is, after the 4th event, hence event CEVT4 can conveniently be used to trigger an interrupt and the CPU can read data from the CAPx registers.

Figure 20-21. Capture Sequence for Absolute Time-stamp and Rising Edge Detect



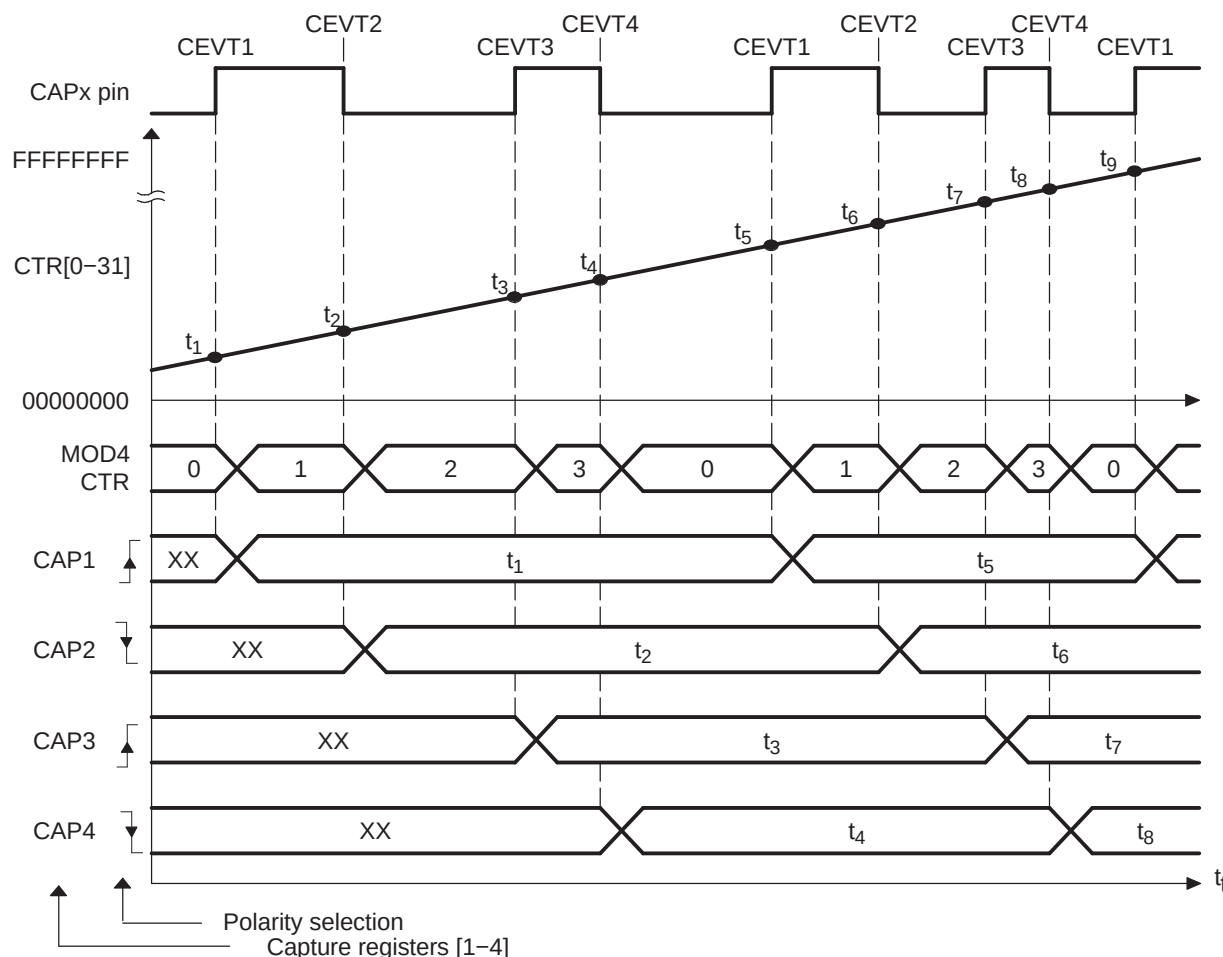
20.6.1.1 Code snippet for CAP mode Absolute Time, Rising Edge Trigger

```
// Code snippet for CAP mode Absolute Time, Rising edge trigger
// Initialization Time
//=====
// ECAP module 1 config
ECap1Regs.ECCTL1.bit.CAP1POL = EC_RISING;
ECap1Regs.ECCTL1.bit.CAP2POL = EC_RISING;
ECap1Regs.ECCTL1.bit.CAP3POL = EC_RISING;
ECap1Regs.ECCTL1.bit.CAP4POL = EC_RISING;
ECap1Regs.ECCTL1.bit.CTRRST1 = EC_ABS_MODE;
ECap1Regs.ECCTL1.bit.CTRRST2 = EC_ABS_MODE;
ECap1Regs.ECCTL1.bit.CTRRST3 = EC_ABS_MODE;
ECap1Regs.ECCTL1.bit.CTRRST4 = EC_ABS_MODE;
ECap1Regs.ECCTL1.bit.CAPLDEN = EC_ENABLE;
ECap1Regs.ECCTL1.bit.PRESCALE = EC_DIV1;
ECap1Regs.ECCTL2.bit.CAP_APWM = EC_CAP_MODE;
ECap1Regs.ECCTL2.bit.CONT_ONESHT = EC_CONTINUOUS;
ECap1Regs.ECCTL2.bit.SYNCO_SEL = EC_SYNCO_DIS;
ECap1Regs.ECCTL2.bit.SYNCI_EN = EC_DISABLE;
ECap1Regs.ECCTL2.bit.TSCTRSTOP = EC_RUN; // Allow TSCTR to run
// Run Time ( e.g. CEVT4 triggered ISR call)
//=====
TSt1 = ECap1Regs.CAP1; // Fetch Time-Stamp captured at t1
TSt2 = ECap1Regs.CAP2; // Fetch Time-Stamp captured at t2
TSt3 = ECap1Regs.CAP3; // Fetch Time-Stamp captured at t3
TSt4 = ECap1Regs.CAP4; // Fetch Time-Stamp captured at t4
Period1 = TSt2-TSt1; // Calculate 1st period
Period2 = TSt3-TSt2; // Calculate 2nd period
Period3 = TSt4-TSt3; // Calculate 3rd period
```

20.6.2 Example 2 - Absolute Time-Stamp Operation Rising and Falling Edge Trigger

In Figure 20-22 the eCAP operating mode is almost the same as in the previous section except capture events are qualified as either rising or falling edge, this now gives both period and duty cycle information, that is, $\text{Period1} = t_3 - t_1$, $\text{Period2} = t_5 - t_3$, and so on; $\text{Duty Cycle1 (on-time \%)} = (t_2 - t_1) / \text{Period1} \times 100\%$, and so on; $\text{Duty Cycle1 (off-time \%)} = (t_3 - t_2) / \text{Period1} \times 100\%$, and so on.

Figure 20-22. Capture Sequence for Absolute Time-stamp With Rising and Falling Edge Detect



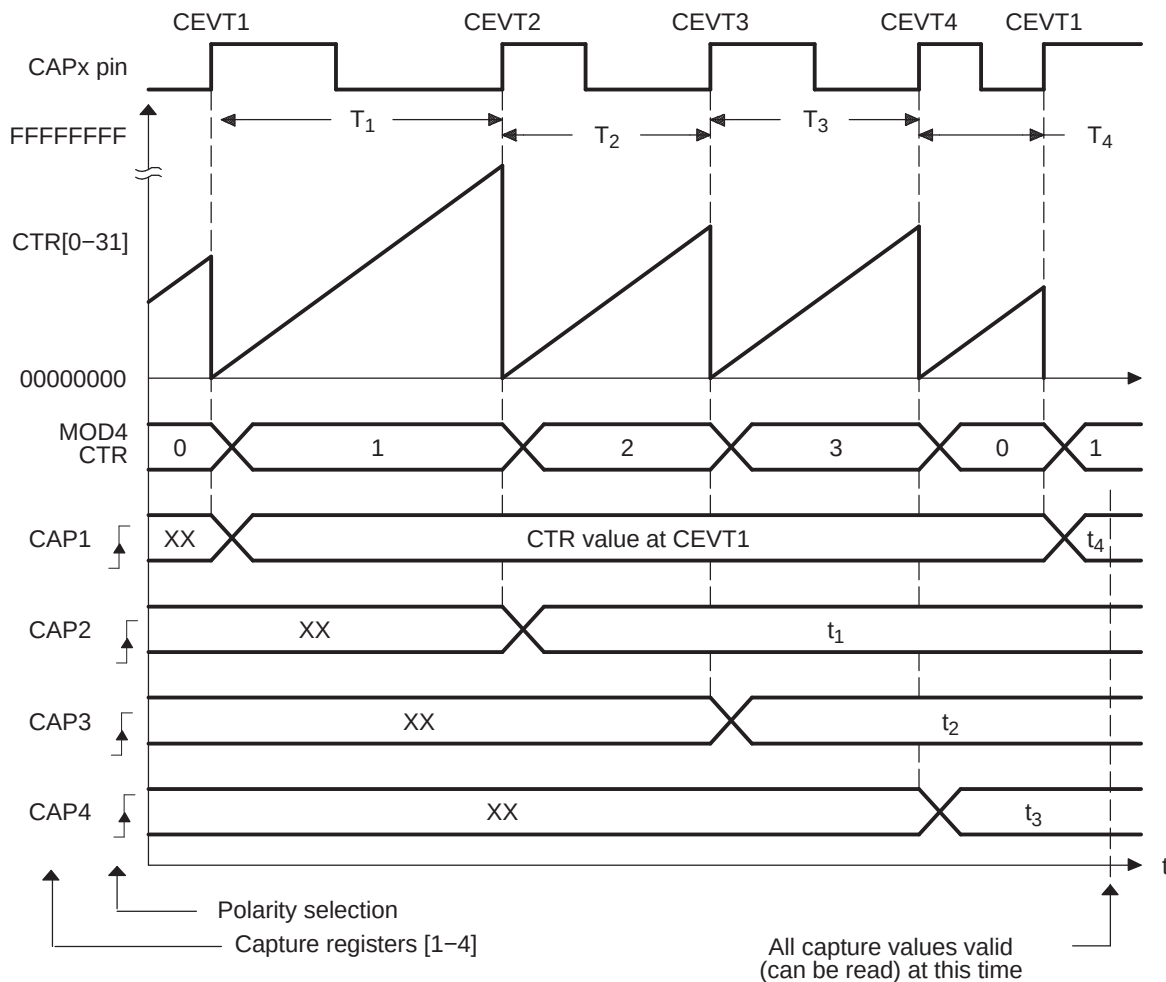
20.6.2.1 Code snippet for CAP mode Absolute Time, Rising & Falling Edge Triggers

```
// Code snippet for CAP mode Absolute Time, Rising & Falling edge triggers
// Initialization Time
//=====
// ECAP module 1 config
ECap1Regs.ECCTL1.bit.CAP1POL = EC_RISING;
ECap1Regs.ECCTL1.bit.CAP2POL = EC_FALLING;
ECap1Regs.ECCTL1.bit.CAP3POL = EC_RISING;
ECap1Regs.ECCTL1.bit.CAP4POL = EC_FALLING;
ECap1Regs.ECCTL1.bit.CTRRST1 = EC_ABS_MODE;
ECap1Regs.ECCTL1.bit.CTRRST2 = EC_ABS_MODE;
ECap1Regs.ECCTL1.bit.CTRRST3 = EC_ABS_MODE;
ECap1Regs.ECCTL1.bit.CTRRST4 = EC_ABS_MODE;
ECap1Regs.ECCTL1.bit.CAPLDEN = EC_ENABLE;
ECap1Regs.ECCTL1.bit.PRESCALE = EC_DIV1;
ECap1Regs.ECCTL2.bit.CAP_APWM = EC_CAP_MODE;
ECap1Regs.ECCTL2.bit.CONT_ONESHT = EC_CONTINUOUS;
ECap1Regs.ECCTL2.bit.SYNCO_SEL = EC_SYNCO_DIS;
ECap1Regs.ECCTL2.bit.SYNCI_EN = EC_DISABLE;
ECap1Regs.ECCTL2.bit.TSCTRSTOP = EC_RUN; // Allow TSCTR to run
// Run Time ( e.g. CEVT4 triggered ISR call)
//=====
TSt1 = ECap1Regs.CAP1; // Fetch Time-Stamp captured at t1
TSt2 = ECap1Regs.CAP2; // Fetch Time-Stamp captured at t2
TSt3 = ECap1Regs.CAP3; // Fetch Time-Stamp captured at t3
TSt4 = ECap1Regs.CAP4; // Fetch Time-Stamp captured at t4
Period1 = TSt3-TSt1; // Calculate 1st period
DutyOnTime1 = TSt2-TSt1; // Calculate On time
DutyOffTime1 = TSt3-TSt2; // Calculate Off time
```

20.6.3 Example 3 - Time Difference (Delta) Operation Rising Edge Trigger

This example [Figure 20-23](#) shows how the eCAP module can be used to collect Delta timing data from pulse train waveforms. Here Continuous Capture mode (TSCTR counts-up without resetting, and Mod4 counter wraps around) is used. In Delta-time mode, TSCTR is Reset back to Zero on every valid event. Here Capture events are qualified as Rising edge only. On an event, TSCTR contents (that is, time-stamp) is captured first, and then TSCTR is reset to Zero. The Mod4 counter then increments to the next state. If TSCTR reaches FFFF FFFFh (that is, maximum value), before the next event, it wraps around to 0000 0000h and continues, a CINTOVF (counter overflow) Flag is set, and an Interrupt (if enabled) occurs. The advantage of Delta-time Mode is that the CAPx contents directly give timing data without the need for CPU calculations, that is, Period1 = T_1 , Period2 = T_2 , and so on. As shown in the diagram, the CEVT1 event is a good trigger point to read the timing data, T_1 , T_2 , T_3 , T_4 are all valid here.

Figure 20-23. Capture Sequence for Delta Mode Time-stamp and Rising Edge Detect



20.6.3.1 Code snippet for CAP mode Delta Time, Rising Edge Trigger

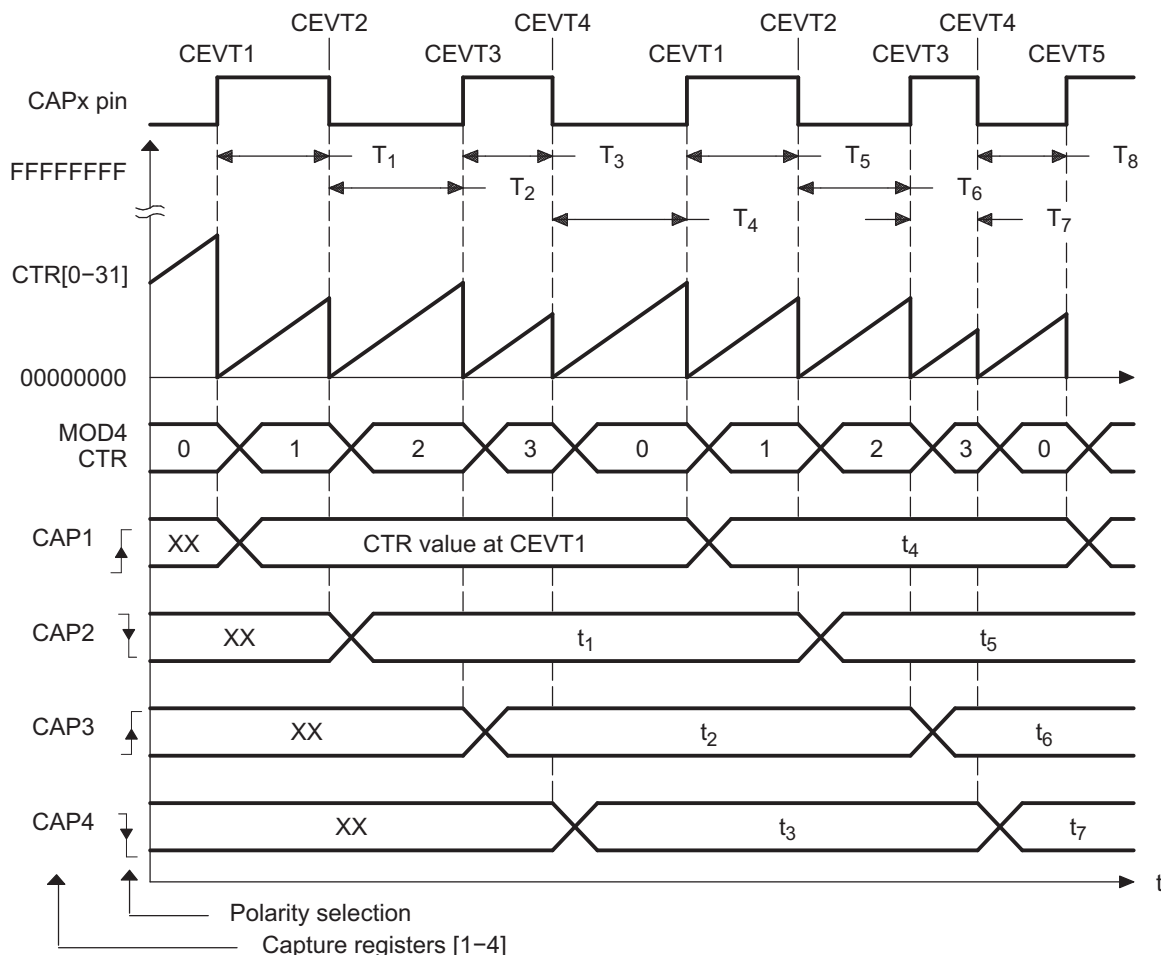
```
// Code snippet for CAP mode Delta Time, Rising edge trigger
// Initialization Time
//=====
// ECAP module 1 config
ECap1Regs.ECCTL1.bit.CAP1POL = EC_RISING;
ECap1Regs.ECCTL1.bit.CAP2POL = EC_RISING;
ECap1Regs.ECCTL1.bit.CAP3POL = EC_RISING;
ECap1Regs.ECCTL1.bit.CAP4POL = EC_RISING;
ECap1Regs.ECCTL1.bit.CTRRST1 = EC_DELTA_MODE;
ECap1Regs.ECCTL1.bit.CTRRST2 = EC_DELTA_MODE;
ECap1Regs.ECCTL1.bit.CTRRST3 = EC_DELTA_MODE;
ECap1Regs.ECCTL1.bit.CTRRST4 = EC_DELTA_MODE;
ECap1Regs.ECCTL1.bit.CAPLDEN = EC_ENABLE;
ECap1Regs.ECCTL1.bit.PRESCALE = EC_DIV1;
ECap1Regs.ECCTL2.bit.CAP_APWM = EC_CAP_MODE;
ECap1Regs.ECCTL2.bit.CONT_ONESHT = EC_CONTINUOUS;
ECap1Regs.ECCTL2.bit.SYNCO_SEL = EC_SYNCO_DIS;
ECap1Regs.ECCTL2.bit.SYNCI_EN = EC_DISABLE;
ECap1Regs.ECCTL2.bit.TSCTRSTOP = EC_RUN; // Allow TSCTR to run
// Run Time ( e.g. CEVT1 triggered ISR call)
//=====
// Note: here Time-stamp directly represents the Period value.
Period4 = ECap1Regs.CAP1; // Fetch Time-Stamp captured at T1
Period1 = ECap1Regs.CAP2; // Fetch Time-Stamp captured at T2
Period2 = ECap1Regs.CAP3; // Fetch Time-Stamp captured at T3
Period3 = ECap1Regs.CAP4; // Fetch Time-Stamp captured at T4
```

20.6.4 Example 4 - Time Difference (Delta) Operation Rising and Falling Edge Trigger

In Figure 20-24 the eCAP operating mode is almost the same as in previous section except Capture events are qualified as either Rising or Falling edge, this now gives both Period and Duty cycle information, that is, $\text{Period1} = T_1 + T_2$, $\text{Period2} = T_3 + T_4$, and so on; $\text{Duty Cycle1 (on-time \%)} = T_1 / \text{Period1} \times 100\%$, and so on; $\text{Duty Cycle1 (off-time \%)} = T_2 / \text{Period1} \times 100\%$, and so on.

During initialization, you must write to the active registers for both period and compare. This will then automatically copy the init values into the shadow values. For subsequent compare updates, during run-time, only the shadow registers must be used.

Figure 20-24. Capture Sequence for Delta Mode Time-stamp With Rising and Falling Edge Detect



20.6.4.1 Code snippet for CAP mode Delta Time, Rising and Falling Edge Triggers

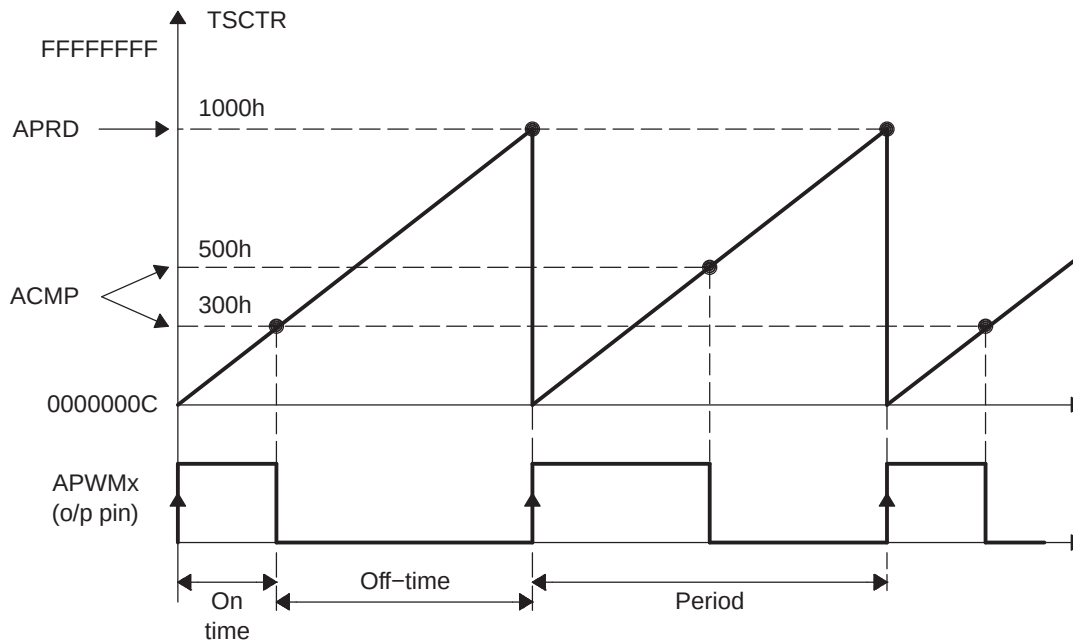
```
// Code snippet for CAP mode Delta Time, Rising and Falling edge triggers
// Initialization Time
//=====
// ECAP module 1 config
ECap1Regs.ECCTL1.bit.CAP1POL = EC_RISING;
ECap1Regs.ECCTL1.bit.CAP2POL = EC_FALLING;
ECap1Regs.ECCTL1.bit.CAP3POL = EC_RISING;
ECap1Regs.ECCTL1.bit.CAP4POL = EC_FALLING;
ECap1Regs.ECCTL1.bit.CTRRST1 = EC_DELTA_MODE;
ECap1Regs.ECCTL1.bit.CTRRST2 = EC_DELTA_MODE;
ECap1Regs.ECCTL1.bit.CTRRST3 = EC_DELTA_MODE;
ECap1Regs.ECCTL1.bit.CTRRST4 = EC_DELTA_MODE;
ECap1Regs.ECCTL1.bit.CAPLDEN = EC_ENABLE;
ECap1Regs.ECCTL1.bit.PRESCALE = EC_DIV1;
ECap1Regs.ECCTL2.bit.CAP_APWM = EC_CAP_MODE;
ECap1Regs.ECCTL2.bit.CONT_ONESHT = EC_CONTINUOUS;
ECap1Regs.ECCTL2.bit.SYNCO_SEL = EC_SYNCO_DIS;
ECap1Regs.ECCTL2.bit.SYNCI_EN = EC_DISABLE;
ECap1Regs.ECCTL2.bit.TSCTRSTOP = EC_RUN; // Allow TSCTR to run
// Run Time ( e.g. CEVT1 triggered ISR call)
//=====
// Note: here Time-stamp directly represents the Duty cycle values.
DutyOnTime1 = ECap1Regs.CAP2; // Fetch Time-Stamp captured at T2
DutyOffTime1 = ECap1Regs.CAP3; // Fetch Time-Stamp captured at T3
DutyOnTime2 = ECap1Regs.CAP4; // Fetch Time-Stamp captured at T4
DutyOffTime2 = ECap1Regs.CAP1; // Fetch Time-Stamp captured at T1
Period1 = DutyOnTime1 + DutyOffTime1;
Period2 = DutyOnTime2 + DutyOffTime2;
```

20.7 Application of the APWM Mode

In this example, the eCAP module is configured to operate as a PWM generator. Here a very simple single channel PWM waveform is generated from output pin APWMx. The PWM polarity is active high, which means that the compare value (CAP2 reg is now a compare register) represents the on-time (high level) of the period. Alternatively, if the APWMPOL bit is configured for active low, then the compare value represents the off-time. Note here values are in hexadecimal (h) notation.

20.7.1 Simple PWM Generation (Independent Channel/s)

Figure 20-25. PWM Waveform Details of APWM Mode Operation



Example 20-1. Code Snippet for APWM Mode

```
// Code snippet for APWM mode Example 1
// Initialization Time
//=====
// ECAP module 1 config
ECap1Regs.CAP1 = 0x1000; // Set period value
ECap1Regs.CTRPHS = 0x0; // make phase zero
ECap1Regs.ECCTL2.bit.CAP_APWM = EC_APWM_MODE;
ECap1Regs.ECCTL2.bit.APWMPOL = EC_ACTV_HI; // Active high
ECap1Regs.ECCTL2.bit.SYNCR_EN = EC_DISABLE; // Synch not used
ECap1Regs.ECCTL2.bit.SYNCR_SEL = EC_SYNCR_DIS; // Synch not used
ECap1Regs.ECCTL2.bit.TSCTRSTOP = EC_RUN; // Allow TSCTR to run
// Run Time (Instant 1, e.g. ISR call)
//=====
ECap1Regs.CAP2 = 0x300; // Set Duty cycle i.e. compare value
// Run Time (Instant 2, e.g. another ISR call)
//=====
ECap1Regs.CAP2 = 0x500; // Set Duty cycle i.e. compare value
```